

Assignment

Zenlynx Technology Private Limited

CIN: U62011OD2024PTC047146

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1. Background

Zenlynx Technology's sustainability platform contains organization-specific data such as emissions, energy, water, waste, and ESG documents.

The objective of this assignment is to build a simple **AI-powered API using Python and RAG** that can answer questions using data belonging to a specific organization.

2. Assignment

Develop a small **Python-based RAG API** that:

- Accepts an `organization_id` and user question.
- Retrieves relevant information from the organization's data/documents.
- Uses RAG to generate an answer.
- Ensures that information from other organizations is not included.
- Returns the answer along with the retrieved source/document.

3. Example

Request:

```
{  
  "organization_id": "org_001",  
  "question": "What was our Scope 1 emission in 2025?"  
}
```

The system should:

User Question

↓

Identify organization

↓

Retrieve relevant organization data

↓

RAG

↓

Generate Answer

↓

Return Answer + Source

Response:

```
{  
  "organization_id": "org_001",  
  "answer": "Scope 1 emissions in 2025 were 1,250 tCO2e.",  
  "sources": [  
    "emission_report_2025.pdf"  
  ]  
}
```

4. Data

Create sample data for at least **2 organizations**.

The data can contain:

- Sustainability/ESG documents
- Scope 1 and Scope 2 emissions
- Energy consumption
- Water and waste data

The candidate may use simple sample documents such as .txt, .pdf, or .md files.

5. Technical Requirements

Use:

- **Python**
- **FastAPI**
- Any suitable vector database/vector store
- Any embedding model
- Any LLM/API
- SQLite/PostgreSQL if required

The candidate should implement a basic RAG pipeline:

Load → Chunk → Embed → Retrieve → Generate

6. Important Requirement

The system must maintain **organization-level data isolation**.

For example, when querying:

`organization_id = org_001`

the RAG system should retrieve information only from `org_001` documents/data.

7. Deliverables

- Working Python API
- Source code / Git repository
- Sample organization data/documents
- README with setup instructions
- Sample API request and response
- Short explanation of the RAG approach

8. Evaluation

The assignment will be evaluated on:

- Python/API implementation
- RAG pipeline
- Retrieval quality
- Correct use of organization context
- Data isolation

- Response quality and source references
- Code structure and readability

The assignment is intended to be a small working prototype. The candidate is not expected to build a production-ready AI system.